

Swiss Institute of Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

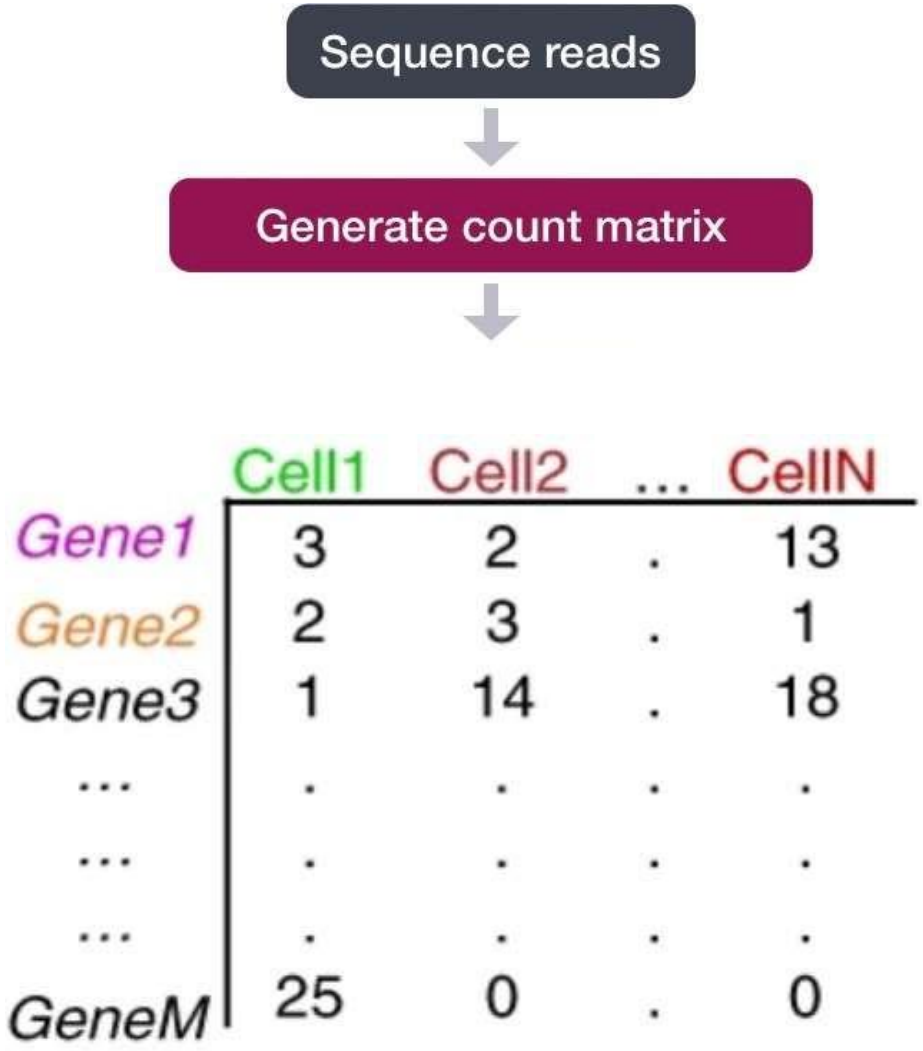
Analysis tools and QC

Deepak Tanwar

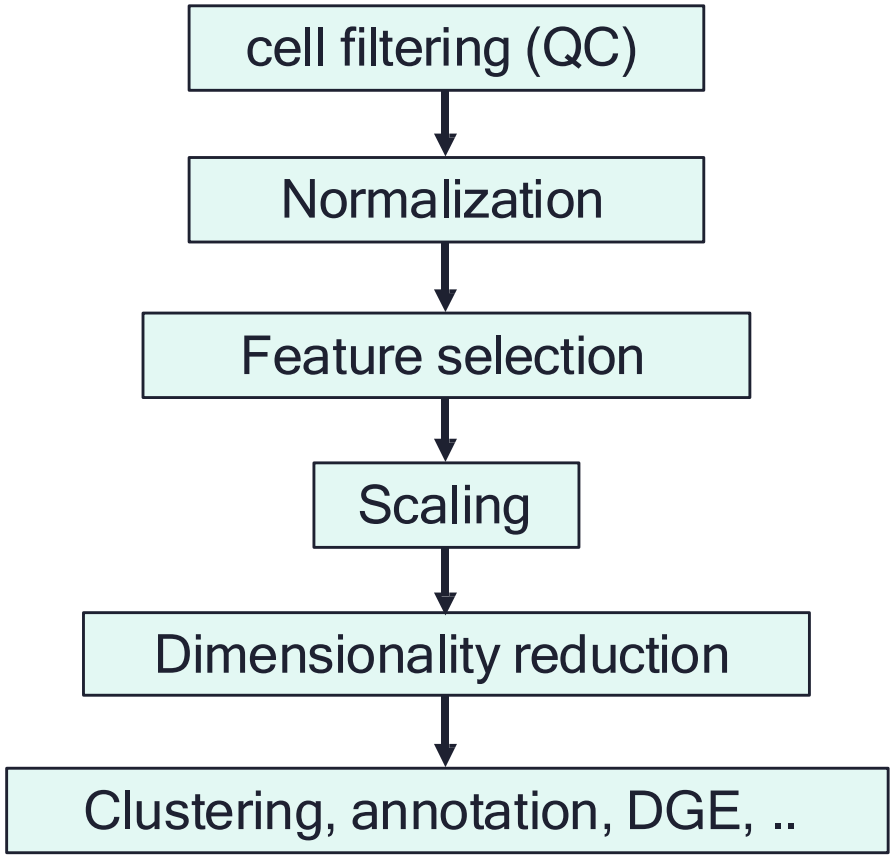
November 11-13, 2026

elixir SWITZERLAND

From FASTQ to counts



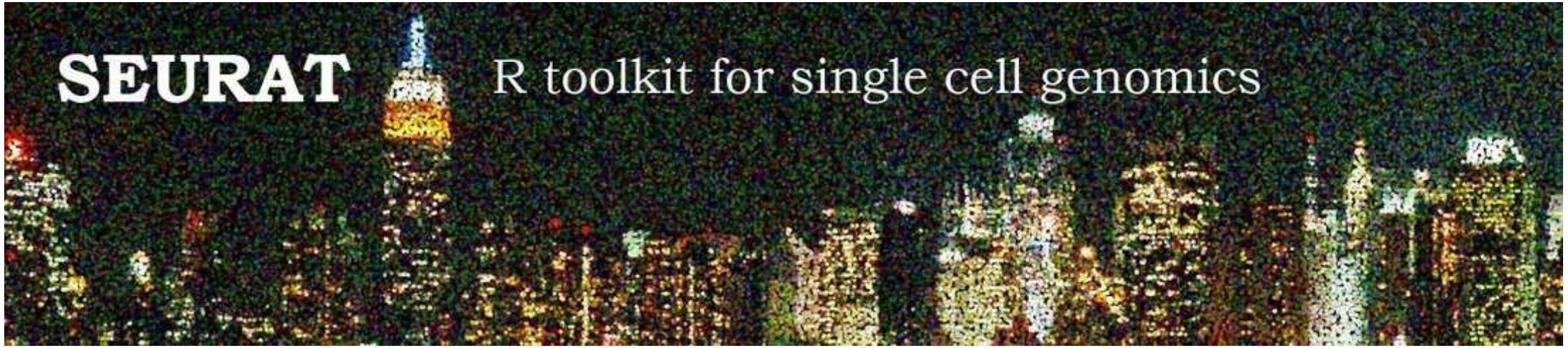
Analysis workflow



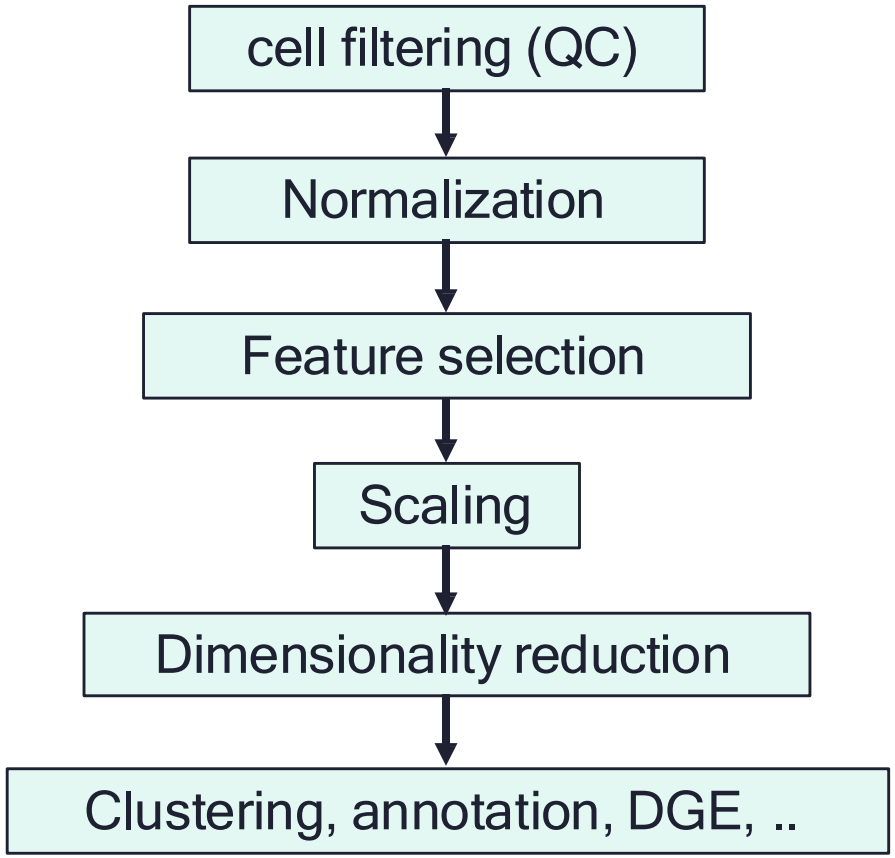
- scanpy (python)
- scater + scran (R, Bioconductor)
- monocle3 (R, beta on github)
- Seurat (R, CRAN)

https://bioconductor.org/books/release/OSCA/

Seurat



Data analysis pipeline overview



cellranger + Seurat

Seurat::NormalizeData()

Seurat::FindVariableFeatures()

Seurat::ScaleData()

Seurat

Seurat, SingleR, clusterProfiler, ..

Quality control of count matrix

Goals:

- To **filter the data to only include true cells that are of high quality**, so that when we cluster our cells it is easier to identify distinct cell type populations
- To **identify any failed samples** and either try to salvage the data or remove from analysis, in addition to, trying to understand why the sample failed

Challenges*:*

Delineating cells that are poor quality from less complex cells

Choosing appropriate thresholds for filtering, so as to keep high quality cells without removing biologically relevant cell types

Cell filtering

Cellranger:

- cell calling (filter against low #UMI)

Manually (e.g. with Seurat):

- **#UMI**: high
- **#detected genes**
- **% mitochondrial UMI**
- **% ribosomal UMI**
- **Relationships between variables**

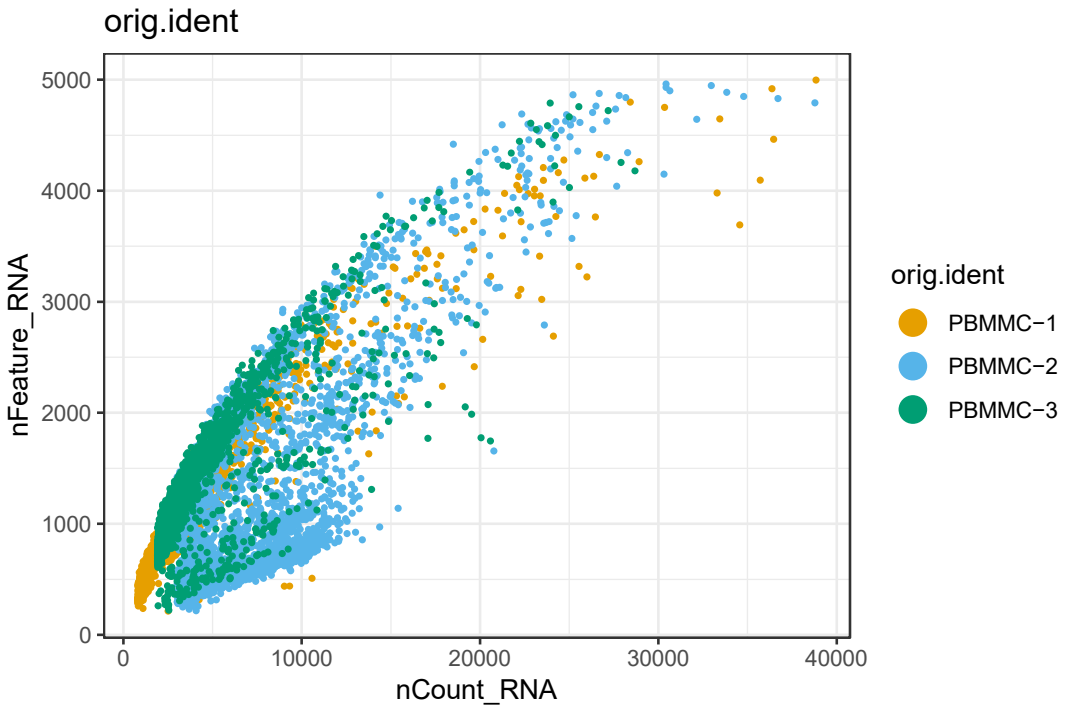
#UMI: high → possible doublet

High UMI counts per barcode often indicate **doublets/multiplets** (two or more cells in one droplet).

Why?

Each cell has a biologically plausible UMI range. Outliers above this suggest merged transcriptomes.

Common threshold: Top 1–5% of UMI distribution or > 2–3× median UMI count



#Detected genes

Low gene count: Empty droplets, damaged cells, or failed library prep.

High gene count: Doublets (two cell types → more genes detected).

Common thresholds:

- Minimum: 200–500 genes/cell
- Maximum: Often tied to UMI count (e.g., cells with >6,000 genes may be doublets in 10x data).

% Mitochondrial UMI: dying cells

High mitochondrial content = cell stress, apoptosis, or membrane damage.

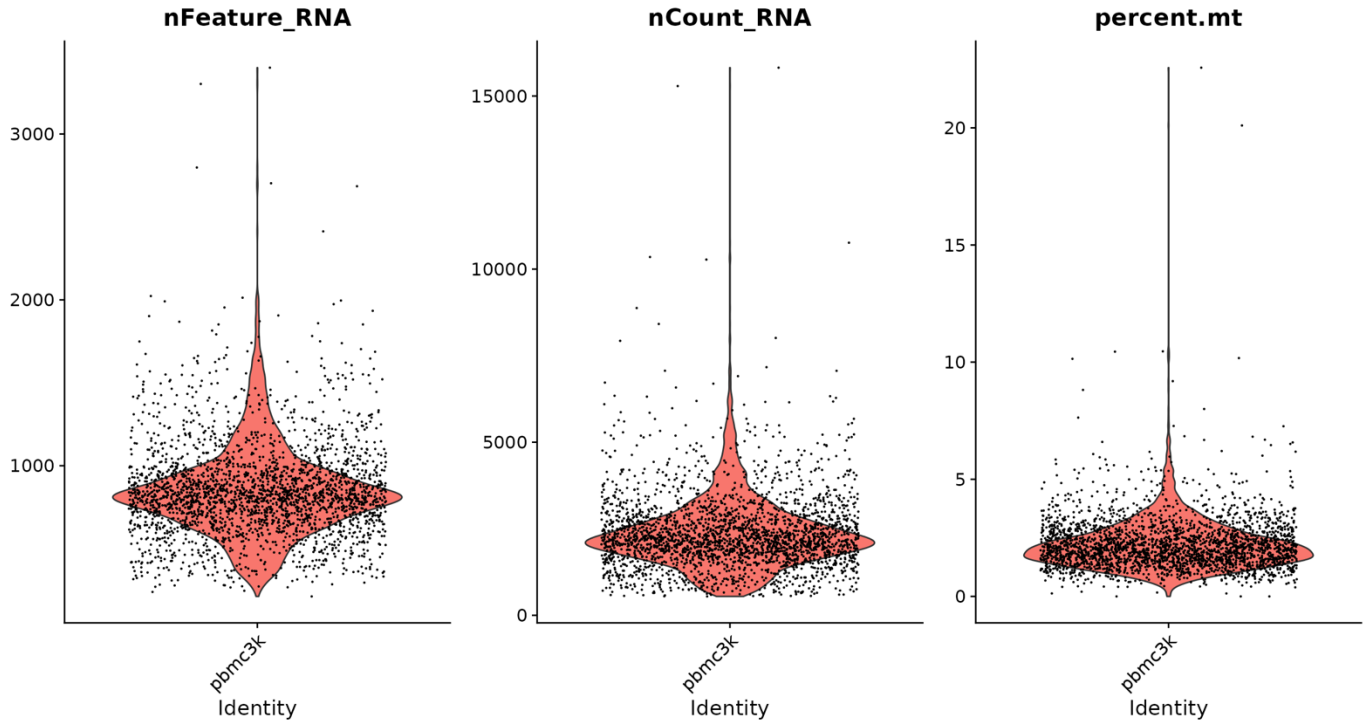
Cytoplasmic mRNA degrades faster than mitochondrial RNA → %MT increases.

Common threshold: Usually <5–20%, depending on tissue (e.g., muscle has naturally high MT).

% Ribosomal UMI

High ribosomal RNA: Technical artifacts (incomplete poly-A selection).

Stress response or rapidly dividing cells.



Relationships between variables

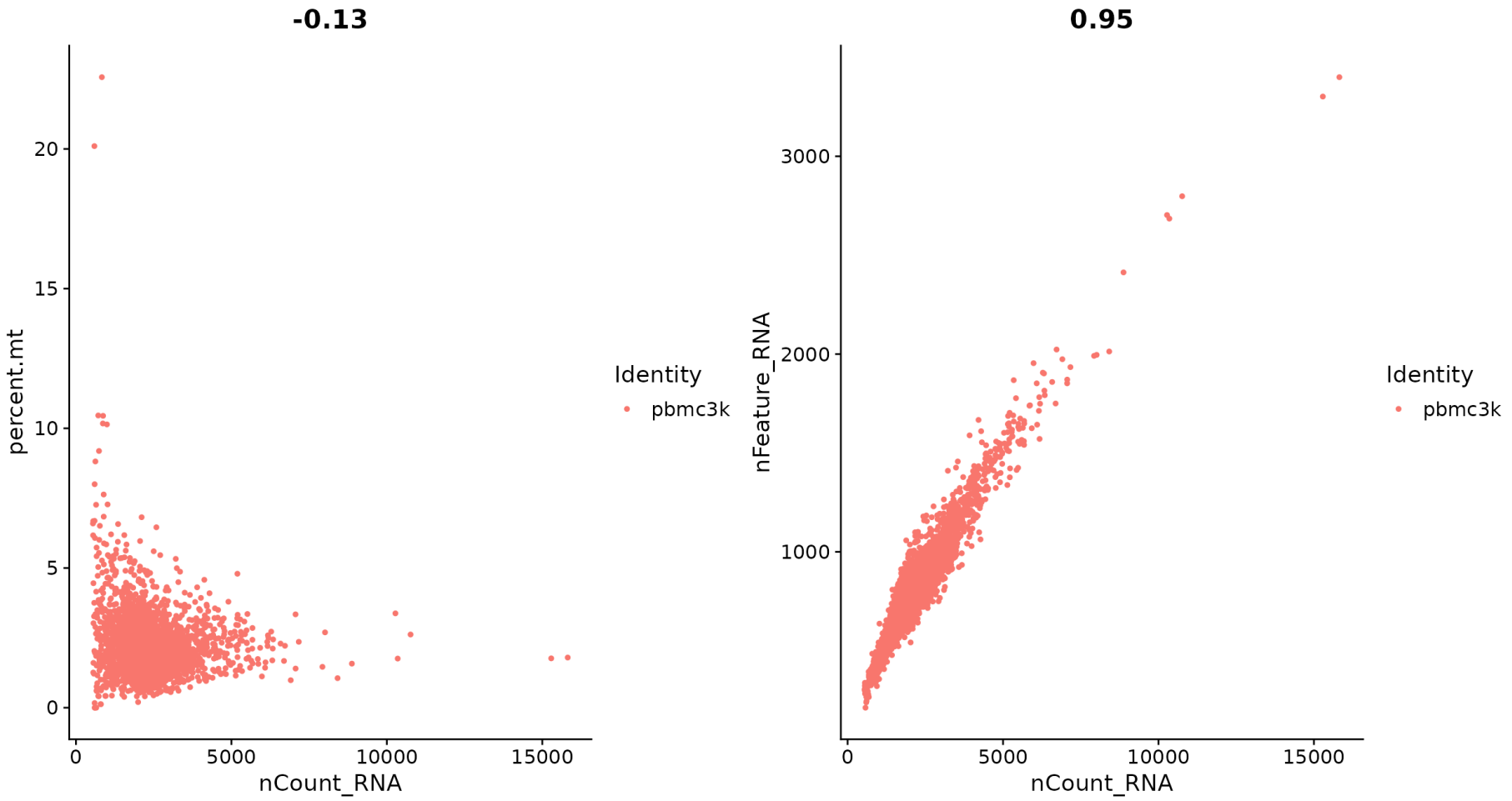
Metrics are not independent:

- High UMI ↔ High gene count
- High %MT ↔ Low UMI/gene count (dying cells lose RNA)

Best practice: Use scatter plots to set adaptive thresholds

Plot	What to Look For	What It Means
#UMI vs #Genes	Linear trend + outliers	Normal cells follow a line. Outliers = doublets
#Genes vs %MT	Low-gene, high-%MT cloud	Dying or empty droplets

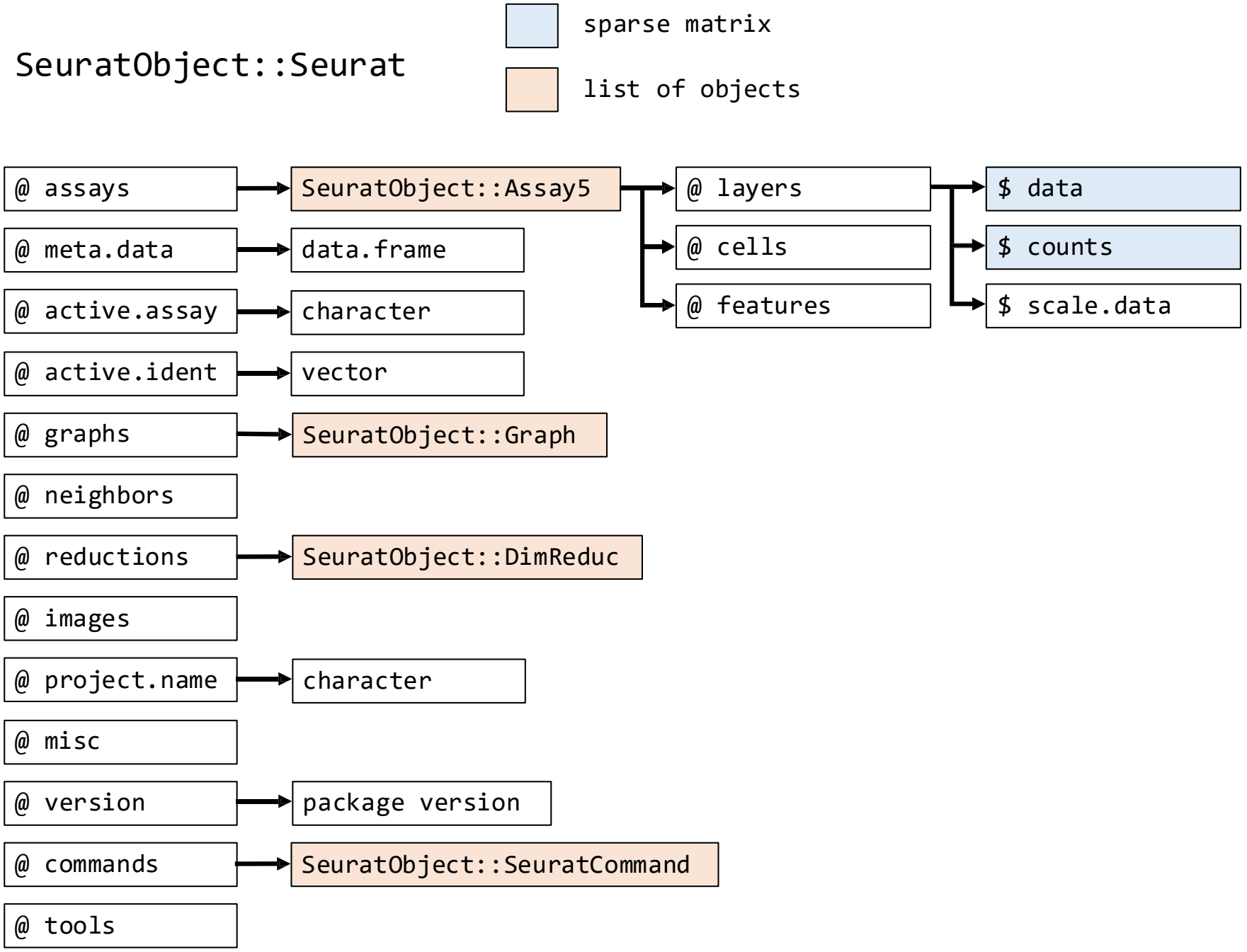
Relationships between variables



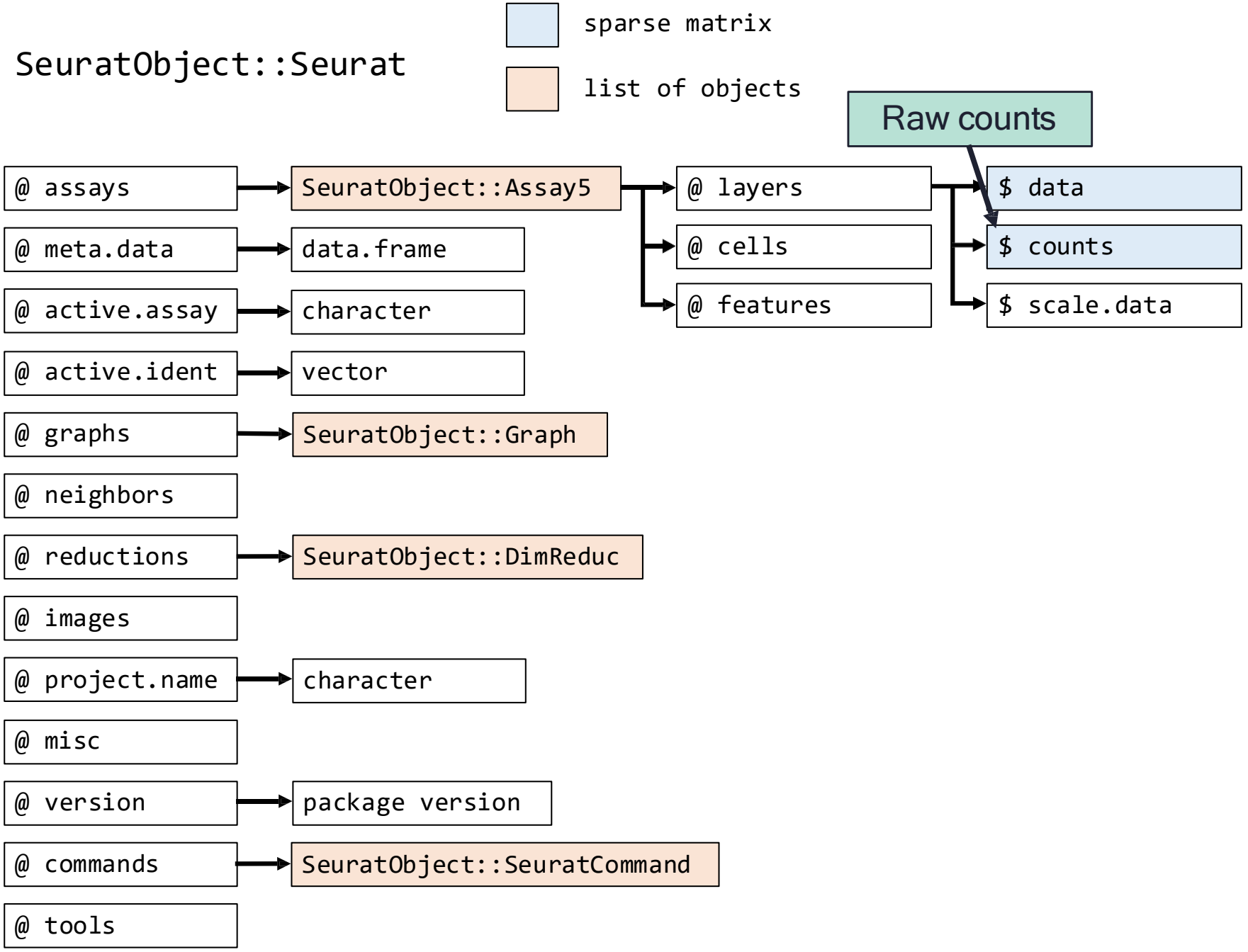
Summary of QC filtering

Metric	Low Value Indicates	High Value Indicates	Typical Action
nCount_RNA (UMI)	Empty droplet	Doublet	Filter extremes
nFeature_RNA (genes)	Low-quality cell	Doublet	Filter low & high
% Mitochondrial	—	Dying/stressed cell	Filter high (>10%)
% Ribosomal	—	Technical bias or stress	Flag or filter

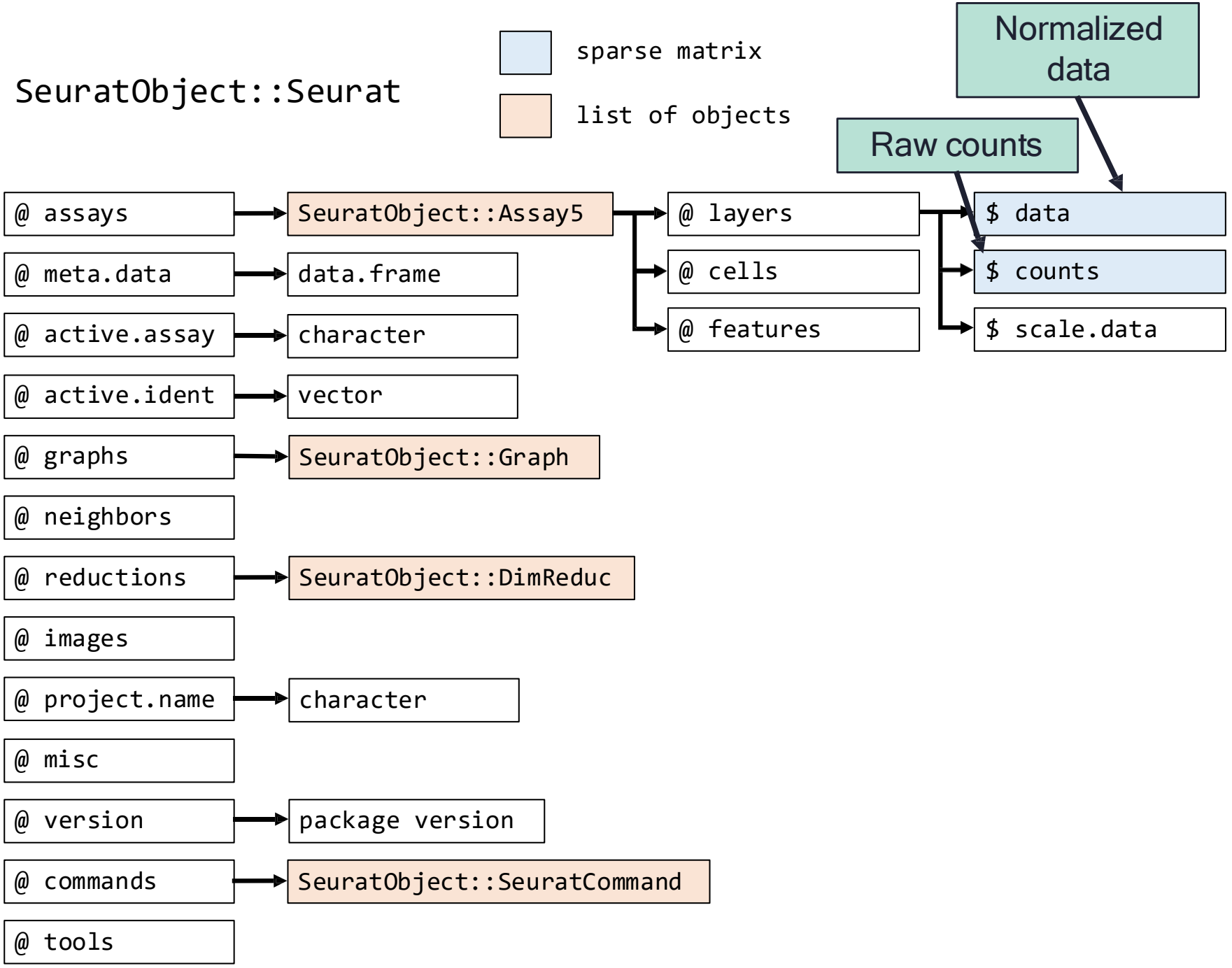
Data structure for the course



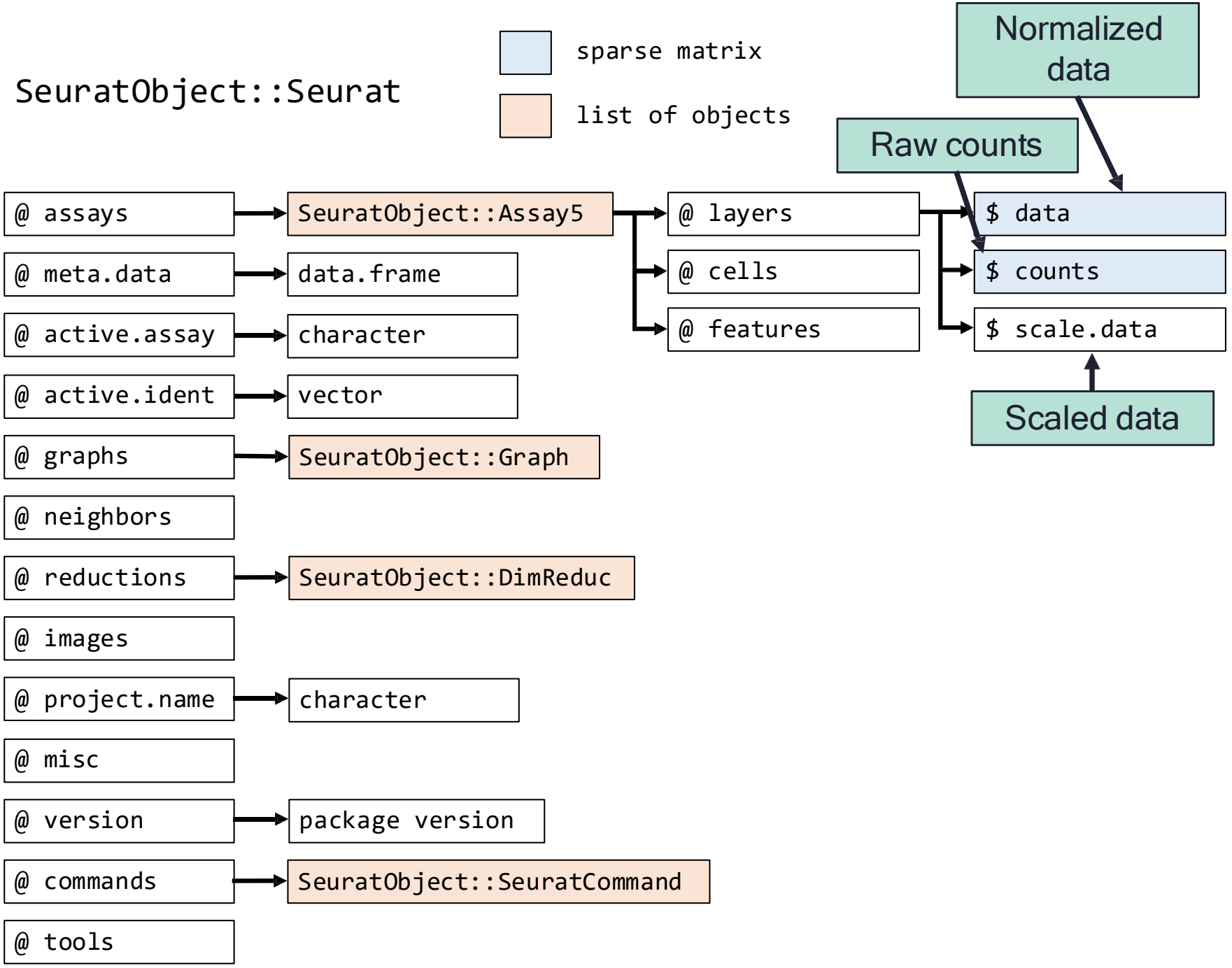
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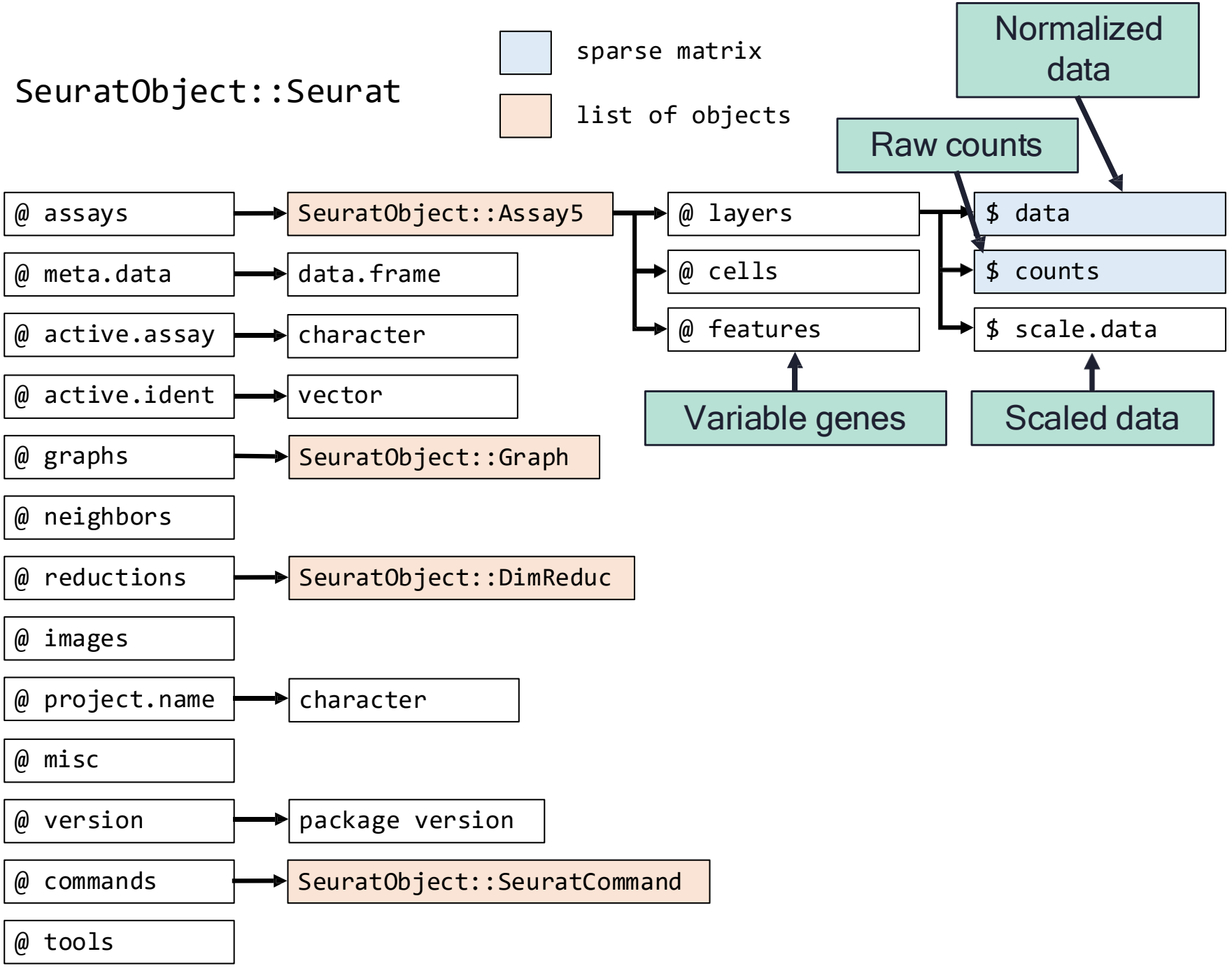
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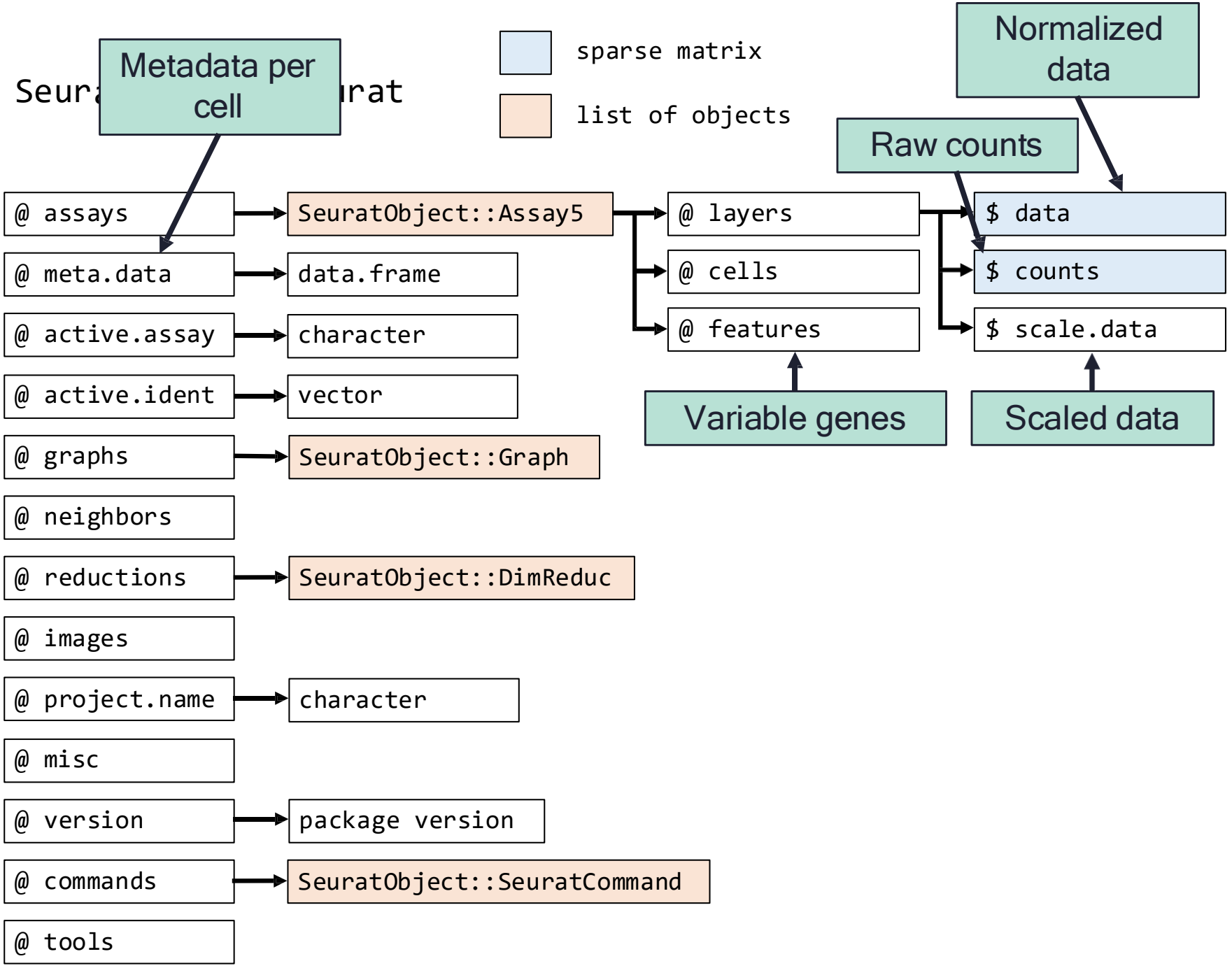
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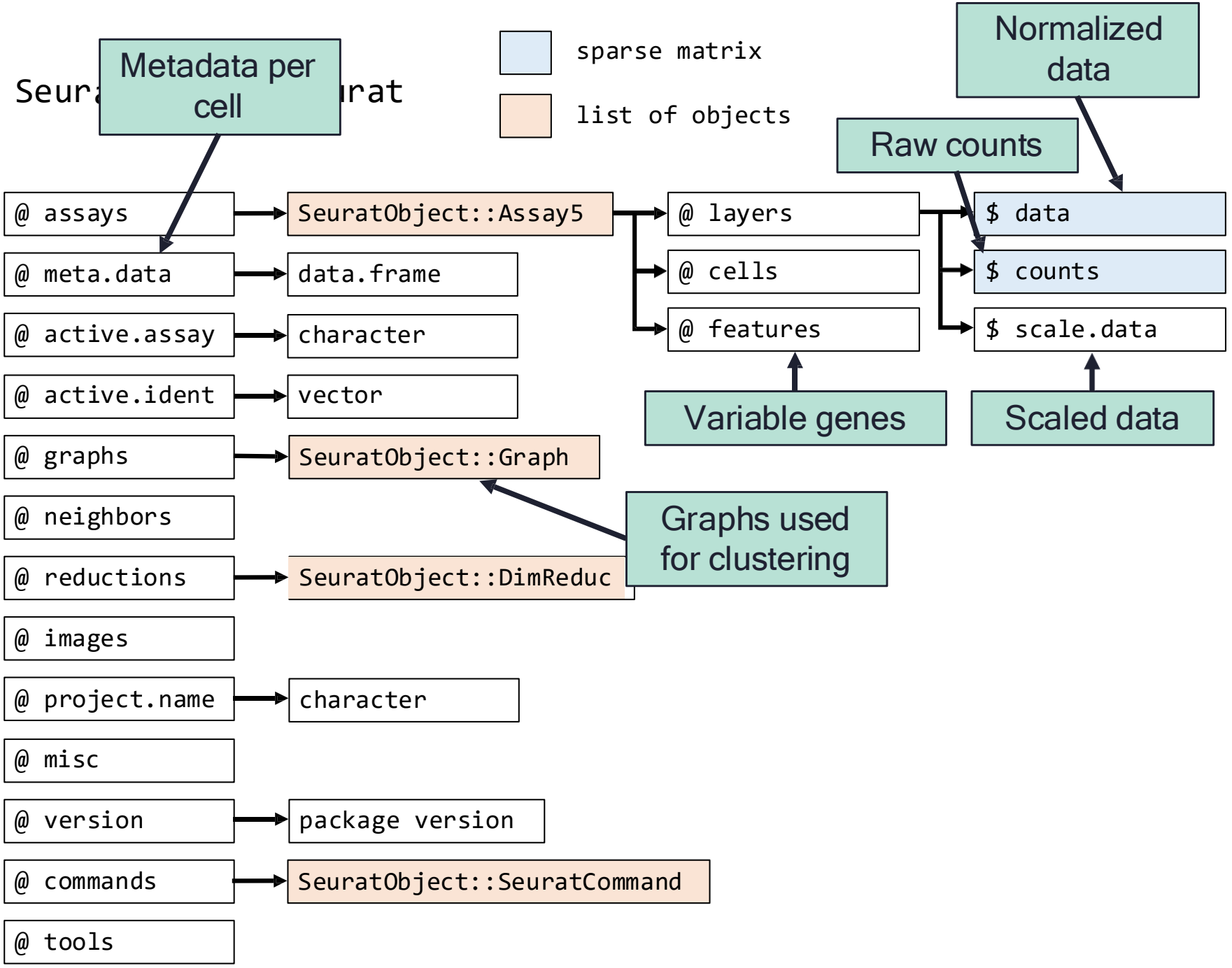
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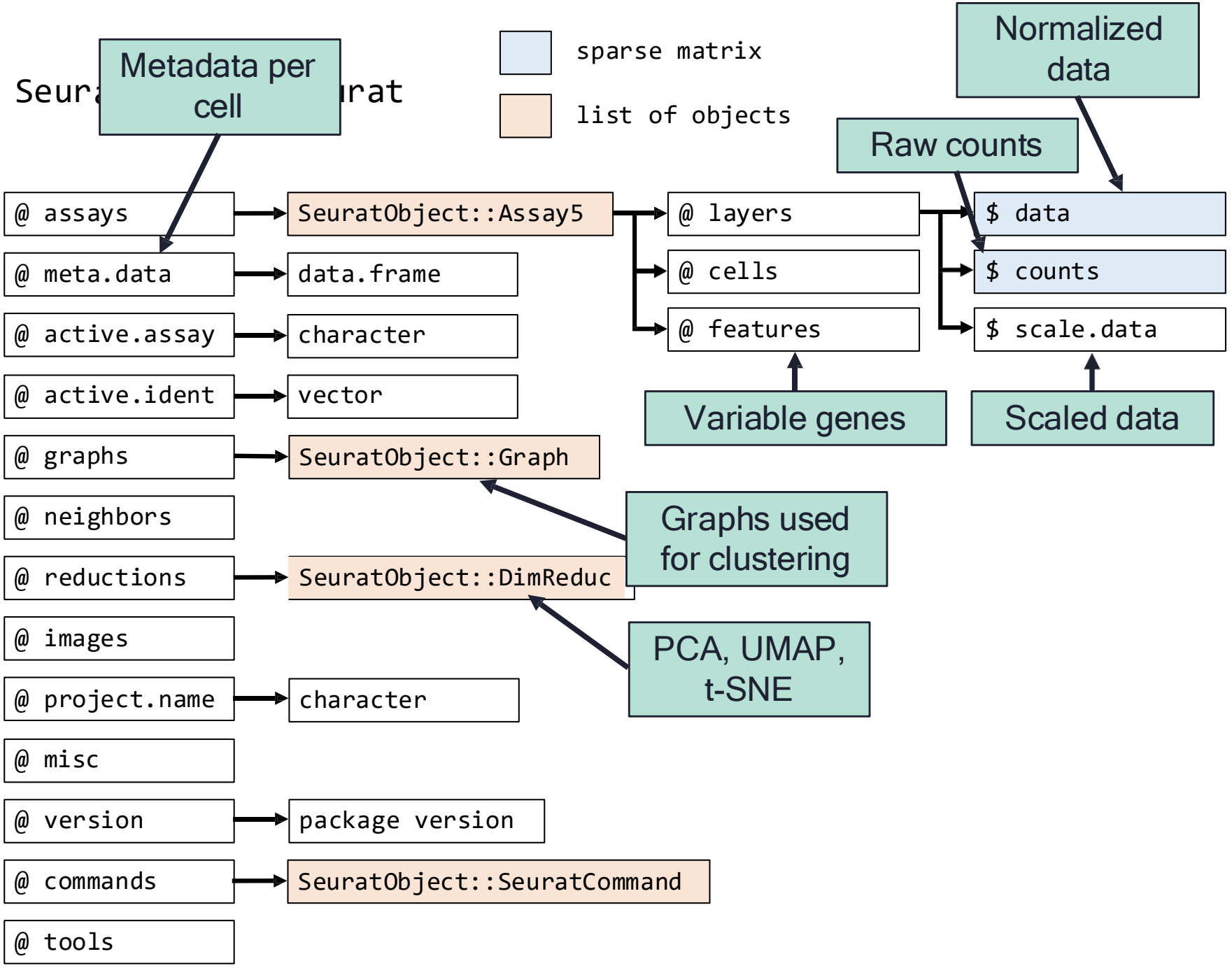
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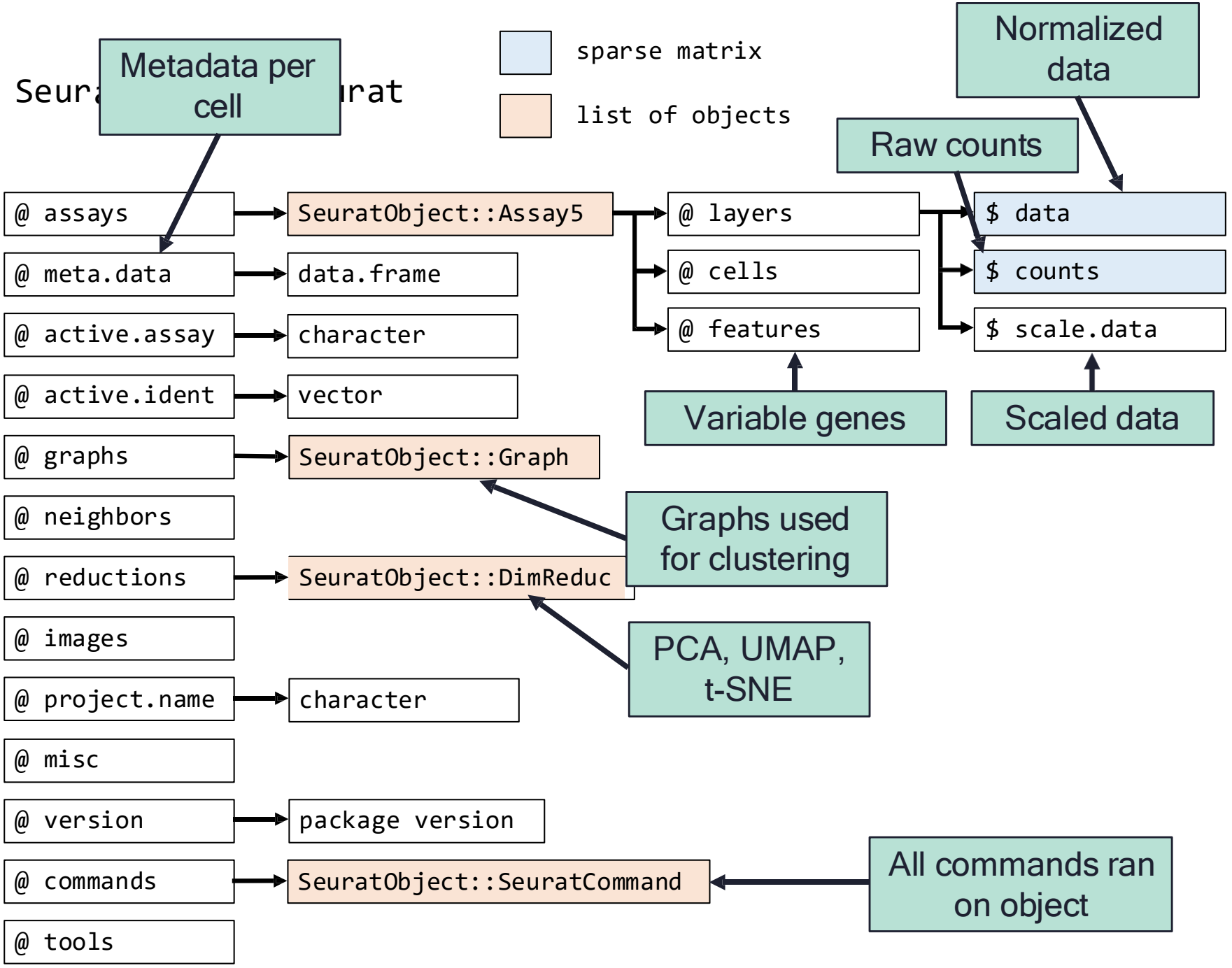
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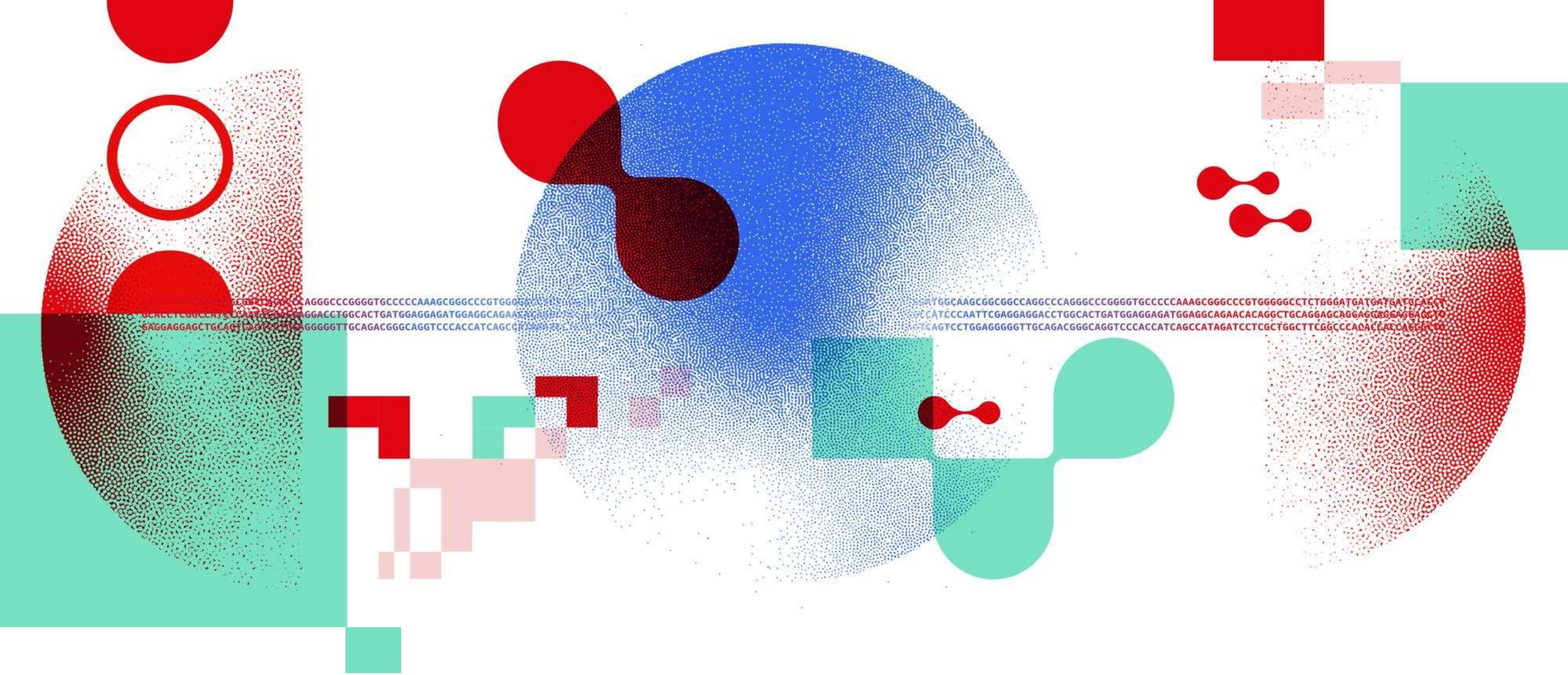


Data structure for the course



Data structure for the course





Thank you

DATA SCIENTISTS FOR LIFE

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